Bachelor's Thesis

From freedom of speech to the U.S. Capitol riots: Topic evolution of content on Parler.

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Abstract

In the context of the 2020 U.S. presidential election, broad interpretations of freedom of speech, guaranteed by the First Amendment of the Constitution, led to unprecedented events. The aim of this thesis is to analyze the relation between posts and comments on Parler, a social media platform allowing its users to express their views unfiltered, and the election timeline. For this purpose, BERTopic, an algorithm implementing the state-of-the-art BERT model for simple and dynamic topic modelling, was performed on four samples extracted from a public dataset of Parler posts and comments. The results revealed that the 2020 U.S. presidential election was the main discussion subject on the platform. According to the topic analysis, Parler user base consisted of supporters of former president Donald Trump, extremists, and conspiracy theorists to a large extent.

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1 Introduction

Throughout time, the urge to interact with each other and the desire to live in a community have been proven to be two of the fundamental needs of human beings. Communication opened up many opportunities, and only after people mastered this art have they become the dominant species on Earth. However, when exceeding certain limits of respect or when guided by leaders who pursued their own interests, communication can lead to the deterioration of the relations between individuals, endangering certain groups and creating conflicts. Social networks have been developed to facilitate conversation between people, and to give them the chance to interact with their peers from everywhere in the world, often without knowing them beforehand. Based on common interests or preferences, people gathered in groups to discuss a wide range of topics, evolving and improving their lives. Yet, this was not the case every time. Discussions on social networks, like the ones in real life, can hardly be restricted. The differences observed in the online environment are the magnitude of the exchange of views and the serious consequences this large-scale communication method can bring. While most large social networks like Twitter take measures to limit violence, harassment, terrorism, misinformation, hate speech, and adult content (*The Twitter rules: safety, privacy, authenticity, and more*, 2022) which proved to have caused problems in the past, smaller platforms do not. Some did simply not reach the development stage of imposing such clear conditions regarding the limits of communication on their platforms. Others desire to appeal to those groups of people marginalized by the big platforms, taking the risk of bearing the consequences of the inappropriate behaviors of their users. By examining a case study, this thesis aims to analyze what significant implications can follow from free discussions on a social network whose rules are not considerably strict. Studying the main topics from posts and comments on the online social network Parler, which promotes itself as "The premier global free speech platform." (*Parler - Where Free Speech Thrives*, 2022), the researchers desire to find out how the discussions on the platform correlate to the United States (U.S.) presidential elections from 2020.

1.1 Research Questions

Hence, three research questions have been defined:

1. Who were the users of the social network Parler and what main topics were they discussing during the (last stage of the) 2020 U.S. presidential elections?

2. Were the topics discussed on Parler in line with the meaning of the democratic ideal of freedom of speech guaranteed by the First Amendment of the Constitution of the U.S.?

3. How were the topics discussed on Parler related to the events of the U.S. presidential elections 2020 and the Capitol riots from 6 January 2021?

In order for those questions to be answered impartially, this research aims to examine Parler discussions by applying a machine learning algorithm for topic modelling to a large dataset of Parler posts and comments from the beginning of November 2020 until the

platform's shutdown in January 2021. The goal of the researchers is to observe the effects of free speech by using a data-driven approach.

1.2 U.S. Presidential Elections 2020

To analyze the connection between the posts and comments made public on the online social network Parler and the concrete events that occurred in the U.S. from November 2020 until January 2021, it is required to comprehend the socio-political context from that time period. One first step is to understand the U.S. presidential election system and to create an overview of the actual circumstances.

The president and the vice president of the U.S. are not elected directly by popular vote, but they are chosen through the Electoral College Process (*Presidential Election Process*, 2022). For the 59th presidential election in the U.S., that has taken place in 2020 in the middle of the COVID-19 pandemic, then-president Donald Trump was running for the Republicans, while the candidate representing the democratic party was the former vice president Joe Biden. The key dates of the electoral process were: November 3, 2020, December 14, 2020, and January 6, 2021. On the general election day, November 3, 2020, the citizens of the U.S. over 18 years voted for their preferred candidate. However, the popular vote does not directly determine the next president, but the result only gives the winning candidate's State political party the right to choose the electors representing the entire State (*Electoral College Timeline of Events*, 2022). Because of the pandemic, a large number of ballots have been received early and by mail, which led to delays in counting and reporting. For that reason, the official results of the popular vote were released only three days later. They revealed that Joe Biden received a percentage of 51.31% and Donald Trump 46.86% of the general election votes (*Official 2020 Presidential General Election Results*, 2022). This caused dissatisfaction and allegations of voter fraud among Trump supporters. Then, 14 December 2020 marked the day the electors voted in each State, votes which were officially counted by the Congress of the U.S. on January 6, 2021. The results showed that Biden received 306 and Trump 232 out of the 538 electoral votes. As the number a candidate needed to win the election was 270 electoral votes, Biden won (*Official 2020 Presidential General Election Results*, 2022). On January 6, 2021, protesters gathered in Washington, D.C. to show their disapproval of Donald Trump's loss, and broke into the Capitol in an attempt to overturn the result of the election (*2021 United States Capitol Attack - Wikipedia*, 2022). The riots only managed to delay the Electoral College vote count by one day. Vice President Mike Pence, as President of the Senate, announced the results on 7 January 2021 and declared that Joe Biden and Kamala Harris have been elected as the next President and Vice President of the U.S. The same day, Donald Trump posted a concession video on his private Twitter account, promising to cooperate in the transition of power (*Trump finally concedes election — CBC News*, 2021). Following the announcement of the President of the Senate, Joe Biden was inaugurated as the 46th president of the U.S. on January 20, 2021.

2 Theoretical Foundation

For an in-depth understanding of the research topic of this thesis, defining the following terms: freedom of speech, the U.S. Capitol riots, Paler, and topic modelling, is required.

2.1 Freedom of Speech

Although an all-time democratic ideal, freedom of speech does not have a law binding definition. The Constitution of the U.S. only declares that the "Congress shall make no law (...) abridging the freedom of speech, or of the press; or the right of the people peaceably to assemble, and to petition the Government for a redress of grievances." (*Constitution of the United States: First Amendment*, 2022). However, the freedom of speech is not limitless, and not every form of speech is protected at all times and under all circumstances. Through a series of landmark cases The Supreme Court of the U.S. has ruled First Amendment freedoms. For example, speech including Advocacy of Illegal Conduct (*Amdt1.2.3.1.1 Advocacy of Illegal Conduct*, 2022), Fighting Words, Hostile Audiences, True Threats (*Amdt1.2.3.2.1 Fighting Words, Hostile Audiences and True Threats*, 2022), Defamation and False Statements (*Amdt1.2.3.3.1 Defamation and False Statements*, 2022), Invasions of Privacy (*Amdt1.2.3.3.2.1 Invasions of Privacy*, 2022), Obscenity (*Amdt1.2.3.4.1 Obscenity*, 2022), Child Pornography (*Amdt1.2.3.4.2.2 Child Pornography*, 2022), Public Indecency and Nudity (*Amdt1.2.3.4.2.3 Public Indecency and Nudity*, 2022) is restricted. This means that speech inciting imminent lawless action, communicating serious threats of physical injury to people, or including conspiracy to commit crimes is not protected under the First Amendment. However, private institutions such as online social media companies are not bound by the First Amendment freedoms, but they establish appropriate behavior on their platforms through their own terms of service. On most online social networks, agreeing with those clauses is mandatory before creating an account. So, if a user fails to comply with this formal agreement, the platform has the right to restrict and even ban him (Berard, 2021). For example, the online social platform Medium, founded by Evan Williams, former CEO of Twitter, has clearly stated rules, not allowing threats of violence and incitement, hateful content, harassment, promotion of controversial, suspect, or extreme content, use of pseudoscience, misinformation, or other content that is contrary to public health or safety, pornographic images or videos, exploitation of minors, promotion and glorification of self-harm and collection of personal information (*Medium Rules*, 2022). They also stipulate that, if breaking the rules, the accounts can be suspended, and content removed even without providing a reason or notification.

2.2 U.S. Capitol Riots

On January 6, 2021, the day the U.S. Congress met in a joint session to count the electoral votes of the presidential elections between Donald Trump and Joe Biden, an unprecedented event took place. Reasons like receiving more in-mail ballots, which caused a delay in counting the number of votes during the popular elections, led to allegations of fraud made by former president Donald Trump and his supporters. They were attempting to overturn the elections by filing lawsuits, spreading conspiracy theories, and showing

unwillingness to cooperate in the transition of the official winner of the elections, Joe Biden (*2020 United States presidential election - Wikipedia*, 2022). The former president Donald Trump was very active on one of the major online social media platforms in the U.S., Twitter, having almost 89 million followers by January 8, 2021, when the platform decided to permanently suspend his account 'due to the risk of further incitement of violence' (*Donald Trump and Twitter – 2009 / 2021 analysis*, 2022). On the morning of January 6, 2021, he posted on Twitter and urged his vice-president Mike Pence, who was also the President of the Senate to declare the elections illegitimate. At noon Donald Trump began a speech in front of his supporters gathered for a protest in Washington DC and started encouraging them to converge on the Capitol (*Capitol riots timeline: What happened on 6 Jan one year ago?*, 2022). Not even two hours later a mob of 2000-2500 protesters was fighting against the police officers in front of the Capitol, managing to break into the building. They were searching for the members of the Senate who stopped their session just minutes earlier and were evacuated.

The riot failed to overturn the result of the presidential elections, only causing a delay in counting the electoral votes. The U.S. Congress met on January 7, 2021, for a second time, they continued certifying the electoral college votes, and Mike Pence declared Joe Biden winner of the election.

However, the riot did not remain without consequences. Besides the physical damages of the Capitol building, the unsettling death of five people has been witnessed (So, 2021). The U.S. Capitol Police reported that both the U.S. Capitol Police Officers Brian Sicknick and Howie Liebengood lost their lives, and nearly 150 law enforcement officers w(*2021 United States Capitol Attack - Wikipedia*, 2022). Among the long-term repercussions, the rioters faced were criminal charges, investigations into domestic terrorism, and losing their jobs (So, 2021). As of the U.S. Capitol Police Press Release from January 7, 2021, some of the charges the rioters faced were: unlawful entry, assaulting a police officer, resisting arrest, unregistered ammunition or firearm (*U.S. Capitol Police Arrests - January 6, 2021*, 2021).

2.3 Parler

Parler is an American online social network founded in August 2018, promoting itself as "The premier global free speech platform." (*Parler - Where Free Speech Thrives*, 2022). Focusing especially on free speech, Parler is the alternative to large social media platforms such as Twitter and Facebook, which restrict the content posted.

M. Otala et al. (2021) compared Parler to Twitter, in an attempt to study platform migration and political polarization, by looking at U.S. Congress Members' accounts. First, they analyzed the functionalities on the two platforms and found that they are very similar, even if the functions have different names. One of the most important features is posting, implemented on all social networks. In the case of Twitter, the posts are called tweets, while on Parler they are called parleys. In addition, there is the possibility to reshare posts, the name used by Parler being "echos". Parler users also have the opportunity to express their appreciation or disagreement with other people's posts, using the "upvote" and "downvote" functions, similar to "likes" on Twitter. Apart from the functionalities, the researchers also explored the rules imposed by the two platforms and

found substantial differences. In the case of Twitter, there is an exhaustive, clear set of limitations that users must follow, while on Parler there are only two rules stipulated under "Community guidelines".

The most recent version of the "Community Guidelines" is the one amended on 11.02.2021 (*Parler Community Guidelines 11/02/2021*, 2021). Interestingly, the document mentions that only after January 2021 have private companies been held responsible for too broad interpretations of the First Amendment and only when they engage in "state action". However, Parler argues that their "mission has always been to offer a social platform in the spirit of the First Amendment of the United States Constitution, one which empowers the individual to think and share freely". As explained in the document, Parler aims not to delete posts or exclude users except in extreme cases, leaving the freedom to filter what one desires to see to each person active on the platform. For this, Parler implemented mute and block functions, which can be applied to specific users or comments containing terms of choice. Yet, the platform limits content according to two different criteria. Their first rule refers to the exclusion of posts related to crimes, civil torts, intellectual property theft, unlawful acts and child sexual abuse, and of content posted by or on behalf of terrorist organizations. The second principle forbids spam posts and the use of bots.

In the context of the events of January 2021, Parler admitted that some posts on the platform were linked to the assault on the Capitol, but it defended itself by arguing that it had sent multiple warnings to the Federal Bureau of Investigation (Hendrix, 2021). Not only prior to the Capitol attack were Trump supporters detailing the way of action on Parler, but also during the assault was the platform used to post videos and photos with rioters (MacGillis, 2021).

Parler faced serious consequences after the riot on 6 January 2021. It was banned by Apple and Google from their app stores within the next three days and ultimately removed entirely by Amazon from its web-hosting services (Nicas & Alba, 2021). Parler managed to relaunch its website on January 17, 2021, but only as a static page without any of the former Parler functionalities. The restored platform reopened for existing users on February 15, 2021, even though posts before the shutdown were no longer accessible. New sign-ups were possible one week after the restored version relaunched. On May 17, a revised version of the Parler app was launched on the App Store (*Parler - Wikipedia*, 2022).

2.4 Topic Modelling

Due to the rapidly expanding number of texts everywhere, extracting valuable information from unstructured textual data has become challenging. A considerable amount of these texts is produced via online social networks as user-generated content. The discovery of latent subjects in text documents has been the task of topic modelling.

In a paper published by Zhao et. al Zhao et al. (2021) recent advances in one of the most valuable areas of natural language processing (NLP), the topic modelling field, have been analyzed. The researchers define the purpose of a topic model as the discovery of "a set of latent topics, each of which describes an interpretable semantic concept". They made a clear distinction between Bayesian probabilistic topic models (BPTMs) and

neural topic models (NTMs).

Until recently, BPTMs, especially latent Dirichlet allocation (LDA) models, have been the algorithms delivering the best results. However, with the improvement of deep neural networks, topic modelling advanced and NTMs are state-of-the-art algorithms nowadays. One of the most relevant NTM is the Bidirectional Encoder Representations from Transformers (BERT) model. The development of this algorithm began when a group of researchers from Google Brain introduced the transformer architecture (Vaswani et al., 2017). Considerably improving the use of the attention mechanism, this architecture represented a substantial advance in the machine learning field. After only one year, Jacob Devlin and his team from Google AI Language published a paper presenting BERT, the new transformer-based language model, which was stacking encoders on top of each other using a bidirectional transformer architecture (Devlin et al., 2018). This new model represented a real breakthrough for a majority of NLP tasks.

Not only the algorithms have changed, but also the way of training. The paradigm of transfer learning has been applied to NLP. It is no more necessary to train a model from scratch, which usually takes a lot of time and requires high computational power. Better results have been achieved when using an already pre-trained model and fine-tuning it for a specific task. Big artificial intelligence (AI) research laboratories like DeepMind, OpenAI, Facebook AI Research (FAIR), or Google Brain usually publish their models, which are trained on large corpora and implemented using open-source software libraries like TensorFlow (Google Brain) or PyTorch (FAIR). For example, Google's pre-trained BERT model can be fine-tuned with just one additional output layer to create state-of-the-art models for a wide range of NLP tasks, including topic modelling.

3 Methodology

3.1 Parler Dataset

The analysis of this thesis is based on a large open dataset (Aliapoulios, Bevensee, Blackburn, Bradlyn, Cristofaro, et al., 2021) consisting of metadata from 13.25 million Parler user profiles, as well as 183 million Parler posts and comments made by 4 million users from August 2018 until January 2021. The information about the users is stored in 9 newline-delimited JSON (.ndjson) files of almost 7.4 GB in total, while the parleys are saved in 167 .ndjson files, taking a storage space of 144 GB.

The researchers who worked on gathering the data published a paper (Aliapoulios, Bevensee, Blackburn, Bradlyn, De Cristofaro, et al., 2021), providing a basic characterization of the dataset. Their findings reveal that the social media platform has seen substantial influxes of new members following celebrity endorsements, as well as a reaction to the 2020 U.S. presidential election. They report that the growth of Parler is related to Twitter being more strict with the content of the posts, fact-checking Donald Trump's tweets, and banning him from the platform after the attack on the U.S. Capitol. Until June 2020, the volume of posts and comments remains consistent at about 10,000 posts and comments per week, but during the last stage of the 2020 U.S. presidential elections, a significant increase in posts and comments, to 1 million posts/comments per week was reported. Moreover, they demonstrate that topics related to conservative views, as well as conspiracy theories and Donald Trump are among the most disputed on Parler. Although a minority of posts/comments include hashtags, the most common ones are related to the U.S. presidential elections (#trump2020, #maga, and #trump) and the alleged election fraud (#stopthesteal, #voterfraud, and #electionfraud), the QAnon conspiracy theory (#qanon, and #thegreatawakening) indicating that many of Parler users are Trump supporters who debate the 2020 U.S. elections. Another interesting finding shows that a majority of the content submitted on Parler is received positively by its users, being 'upvoted' rather than 'downvoted', which leads to the assumption that most of the users have the same viewpoints.

One of the aspects that should be appreciated is the fact that both the information about posts and comments as well as the metadata from the user's profiles were published unfiltered, allowing for a more accurate analysis. Considering the aim of this research, the thesis only focused on the .ndjson files containing posts and comments, which are called parleys on the platform. These were collected from a random set of a little over four million users chosen from the 13.25 million Parler accounts Aliapoulios et al. gathered profile information from. The resulting dataset contained a total of 183.056 million parleys collected from 01.08.2018 until 11.01.2021. Each of the 167 .ndjson files storing parleys consisted of around 110,000 data entries in the form of key-value pairs. The most important keys amongst the 38 stored were: 'comments' (storing either a numerical value representing the number of comments a post has generated or -1 if the parley was a comment), 'body' (containing the text of the post or comment), 'bodywithurls', 'createdAt' (storing the timestamp the parley was posted at), 'createdAtformatted', 'followers', 'following', 'sensitive', 'upvotes', 'verified', 'reposts', 'downvotes'.

Due to computational power, storage space and time optimization concerns, for the analysis of the thesis four .ndjson files containing parleys information were selected. Be-

cause all the data within the dataset was gathered by crawling, the choice of the files should not affect the result in any significant way. This claim is also supported by the congruent results that were achieved by the multiple samples the researchers experimented with before settling on the four described below.

The sample consisted of the following files:

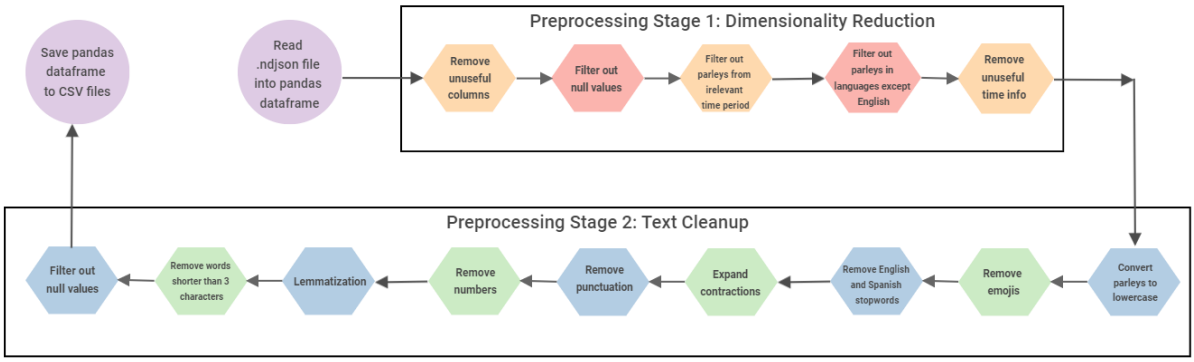
- "parler_data000000000030.ndjson",
- "parler_data000000000031.ndjson",
- "parler_data000000000040.ndjson",
- "parler_data000000000041.ndjson".

Each contains a similar number of records, with the first file having 1,097,921 entries, and the last one 1,095,543.

3.2 Data Preprocessing

In most cases, NLP tasks require text to be preprocessed before training a model, and topic modelling is no exception. Therefore, a thorough selection and cleanup of the Parler dataset containing posts and comments was conducted, in order to help the topic modelling algorithm deliver better results. For this purpose, the Python programming language, the Jupyter Notebook web application and the pandas library (*Pandas - Python Data Analysis Library*, 2022) were used. The process consisted of two main stages: dimensionality reduction and tidying of the textual data, both including several steps. One by one, each of the four .ndjson files chosen for this analysis underwent the same preprocessing methods, illustrated in Figure 1.

Figure 1: Preprocessing pipeline



The first stage of the preprocessing procedure was selecting relevant information from the .ndjson file. The first step was reducing the dimension of the file loaded into a pandas dataframe by deleting columns with irrelevant information for the task given. The only two columns taken into consideration were the one containing the actual text of the parley labeled as [body] and the column referring to the date and time of writing, in the formatted version, labeled as [createdAtFormatted]. Every single initial dataframe had

	Number of parleys (.ndjson file number 30)	Number of parleys (.ndjson file number 41)	Percentage of the initial dataframe
.ndjson file read into pandas dataframe	1,097,921	1,095,543	100%
Dataframe after first filtering of null values (stage 1 of preprocessing)	638,636	637,020	58.16%
Dataframe after filtering of parleys besides the relevant time period (stage 1 of preprocessing)	353,660	353,028	32.22%
Dataframe after filtering of parleys in languages except English (stage 1 of preprocessing)	294,641	294,295	26.85%
Dataframe after second filtering of null values = final complete dataframe (stage 2 of preprocessing)	286,569	286,237	26.11%
November paryles from the final complete dataframe	164,948	164,708	15.03%
December paryles from the final complete dataframe	93,916	93,843	8.56%
January paryles from the final complete dataframe	27,705	27,686	2.53%

Table 1: Data preprocessing

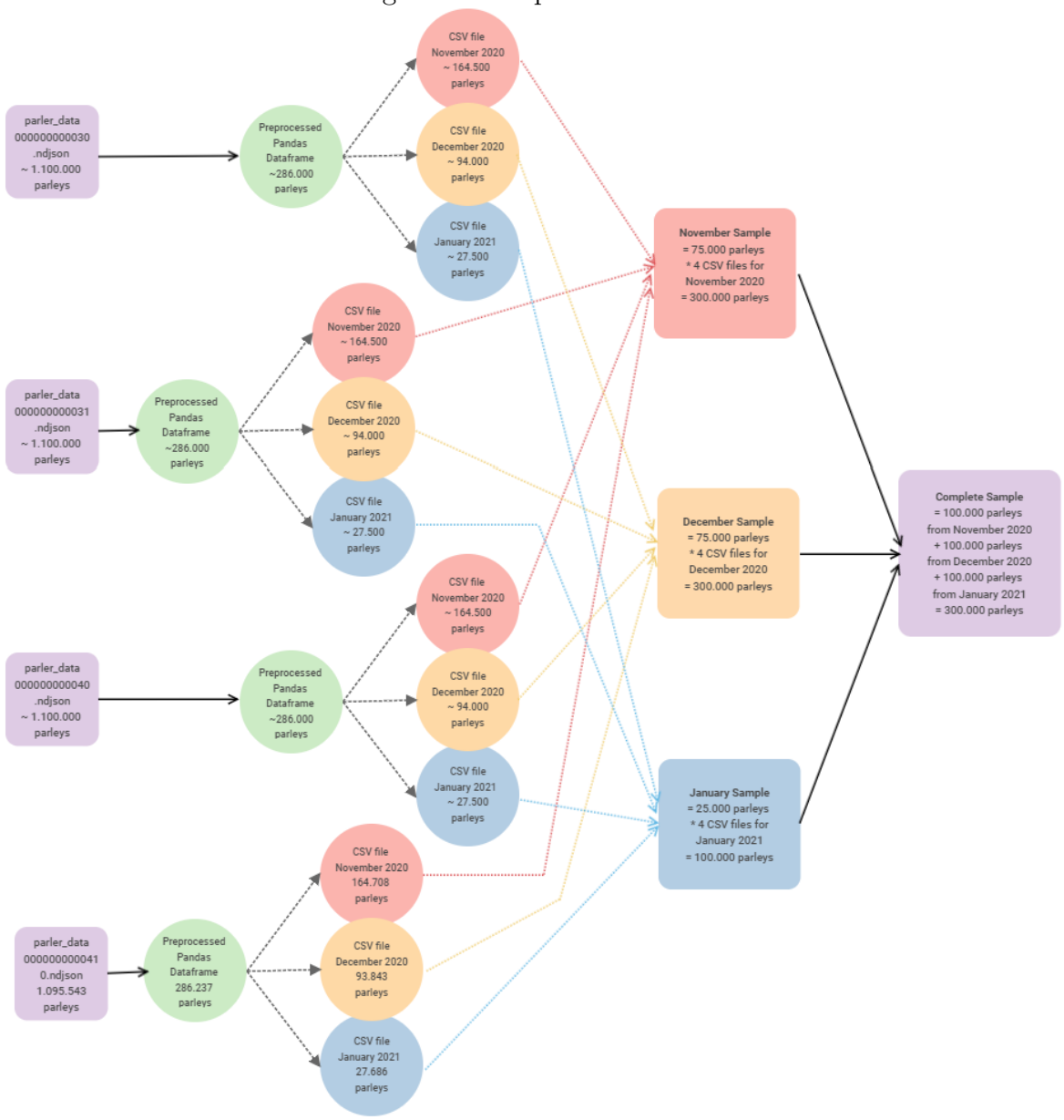
almost 1,100,000 rows and exactly 38 columns (key-value pairs of the .ndjson file) and was reduced to only the two relevant columns. Then, for the dimension reduction of the rows, three operations were performed. First, posts and comments whose [body] contained a null value were filtered out, which resulted in a decrease of the initial dataframe by almost 42%. The explicit dimensions of the .ndjson file number 30 and number 41 which were both chosen subjects of the analysis can be regarded in Table 1. Second, parleys posted beyond the relevant time period were dropped out from the dataframe, causing a further reduction from around 637,000 to only 353,000 rows. As mentioned before, the time period chosen for this research was the last stage of the U.S. presidential elections 2020, from November 2020 until January 2021. The last row dimension reduction operation was filtering out parleys whose text contained mostly non-English words, by using the fasttext language identification model, which can recognize more than 175 languages (*fastText - Language Identification*, 2022). This further decreased the dataframe to less than 300,000 posts and comments, only about 27% of the initial dataframe. From the column storing the timestamp of the parleys, the part specifying the time was deleted, as only the date was useful for splitting the resulting documents. After the first preprocessing stage the dataframe contained 2 columns and about 295,000 posts (exactly 294,295 rows in the case of the last file chosen). Deleting duplicates was not a valid option as the dataset contains a large number of reposts. Because the topic modelling algorithm takes into account the frequency of words, deleting duplicates could have led to poor quality results of the algorithm.

The second stage of the preprocessing work focused on cleaning the parleys and preparing them for the topic modelling algorithm. Working on the [body] column of the dataframe, the first step was simply converting all characters to lowercase. Then,

using the demoji library (*Demoji*, 2022), emojis were accurately found and removed from the text. Afterward, the stopwords library (*Nltk-stopwords: documentation*, 2022) was imported from the nltk corpus package (*Nltk.corpus: documentation*, 2022) providing a list of the most common words from a language. Because those words do not hold descriptive power for a topic, both English and Spanish (as the second most common language) stopwords have been removed from the parleys. The next step was expanding contractions commonly found in informal writings like social media posts and comments by using the contractions library (*Contractions: documentation*, 2022). For example, "you're" has been transformed to "you are". Then, all punctuation characters found in the string library (*String — Common string operations*, 2022) have been deleted. The numbers were also removed with the help of the re regular expression module (*re — regular expression operations*, 2022). Afterward, words shorter than three characters have been deleted, as the probability for those to provide meaningful information in the topic modelling algorithm was low. Finally, after all the steps refining the text of the parleys were completed, filtering out null values was required for a second time. The resulting dataframe had about 287,000 entries, which represented only a little over 26% of the initial number of parleys. This complete dataframe consisting only of clean posts has been divided according to the timestamp into separate dataframes for November 2020, December 2020, and January 2021. These three dataframes were saved as CSV files. For the .ndjson file number 41 the resulting CSV files contained 164,708 parleys for the November sample, 93,843 for the December sample and 27,686 for the January one, which means 15%, 8.5%, and 2.5% of the initial dataframe. The percentages were almost identical for all the four .ndjson files that have been preprocessed.

As displayed in Figure 2, after preprocessing all four .ndjson files taken into consideration for the analysis, 12 CSV files were produced, four for each of the three months regarded (November, December, January). From these files, four samples were created and forwarded one by one as input to the topic modelling algorithm. The first sample consisted of posts exclusively from November 2020. It was constructed by choosing a number of 75,000 parleys from each of the four CSV files containing preprocessed parleys from November. This resulted in a November sample size of 300,000 parleys. The process for the creation of the second sample, December 2020, was the same as for the November 2020 one. For the January 2021 sample, only 100,000 parleys (25,000 from each of the four CSV files containing preprocessed parleys from January) were selected, because Parler was available only until 10 January 2021, a third of the month. The last sample, the complete one, contained 100,000 each of the three month: 100,000 from November 2020, 100,100 from December 2020, and 100,000 from January 2021, 300,000 posts in total.

Figure 2: Sample formation



3.3 BERT using BERTopic

The chosen implementation of BERT in the context of topic modelling for this case study is BERTopic, a "topic modelling technique that leverages transformers and c-TF-IDF to create dense clusters allowing for easily interpretable topics whilst keeping important words in the topic descriptions", developed by the data scientist Maarten Grootendorst (Grootendorst, 2022b). This section aims to explain the three main steps the algorithm is based on and to introduce the technique of dynamic topic modelling.

The first step of the algorithm is embedding the text data. To do so, the BERT sentence transformer "all-MiniLM-L6-v2" model for the English language was chosen, mapping sentences to a 384 dimensional dense vector space (*Sentence-Transformers: all-*

MiniLM-L6-v2, 2022).

The second step is clustering the documents. This requires reducing the dimensionality of the embeddings generated by the BERT transformer. For this task, the Uniform Manifold Approximation and Projection (UMAP) technique for dimension reduction introduced by McInnes et al. (2018) was used. The advantages of UMAP are the high speed of computation compared to other algorithms like t-SNE and better preservation of the local and global structure of the data. However, due to the stochastic nature of UMAP, which makes use of randomness, even multiple executions of the exact same algorithm produce slightly different results. The variations are usually small, but hard to avoid without reducing the performance. For the actual clustering after the dimensionality reduction Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) (Campello et al., 2013) (Campello, 2013), a fast and robust algorithm returning meaningful clusters of varying densities and allowing the identification of outliers was chosen.

The last step is to create a representation for each topic by extracting and reducing topics with class-based term frequency-inverse document frequency (c-TF-IDF), which evaluates how relevant a word is to a document or cluster of documents in this case. Each cluster is considered a document, not a set of documents, and the results of the algorithm are therefore the importance scores for words within a cluster. The most important words in a cluster are simply the descriptions of the topics (Grootendorst, 2022a).

BERTopic is not only able to discover the topics within a corpus, but through a technique called dynamic topic modeling (DTM), the algorithm can analyze the evolution of topics over time. It does so by calculating the topic representation at each timestep without the need to run the complete model several times. For DTM, the first step is fitting a BERTopic model as if the data had no time dimension. Then, the c-TF-IDF representation is calculated for each topic and timestep. This will result in a unique topic representation for each timestep, eliminating the need to generate clusters from embeddings, because they have already been computed (Grootendorst, 2022c).

BERTopic was implemented using the Python programming language, running the algorithm in a Jupyter Notebook in the Google Colab environment. Google Colab was preferred, because it provides access to good quality hardware like GPUs, which is required by transformer models, especially when extracting the sentence embeddings, to speed up computation time. For both faster GPUs and to avoid memory issues when performing the computations on the datasets which were considerable in size, the Google Colab Pro subscription was chosen.

4 Results

As explained in the Methodology section, the case study of this thesis was based on four samples from a dataset containing posts and comments from the Parler social network. The parleys posted during the following distinct time periods: November 2020, December 2020, and January 2021, were analyzed independently since those months were the most relevant for the 2020 U.S. presidential election.

While a total of 300,000 preprocessed parleys were taken into consideration for the November sample and the same number for the December one, the analysis of the January discussions only took 100,000 parleys into account. This smaller sample size has been selected, because the platform was closed on January 10, 2021, due to the incidents in Washington. In addition to the three samples observing the months separately, a fourth was created, containing a total of 300,000 posts, 100,000 parleys from each of the three months. For all four samples, the algorithm was trained with the hyperparameter specifying the number of topics set to 30. This number was chosen after several optimization attempts of both the accuracy of the results and the required execution time. For example, when the number of topics was not specified at all, for a sample of approximately 150,000 parleys, the algorithm found more than 1,000 topics in approximately five hours of training. Although using the fast GPUs provided on Google Colab, both the long execution time and the high number of topics discovered made the analysis unfeasible.

4.1 Topic Descriptions and Categories

The results of the implementation of the BERTopic algorithm for each of the four datasets are displayed in Table 2. As shown in the first column of the table, the 30 topics range from 0 to 29, -1 only referring to the outliers, which should typically be ignored. The second column consists of the results from the complete sample with 100,000 parleys from each month as specified above. The third column contains the topics discovered by the algorithm in the sample with parleys from November 2020, the forth one from December 2020, and the last consists of the topics in the sample with data from January 2021. In some cases, the four words that define the topics in Table 2 do not hold enough descriptive power, even though the have been selected by the algorithm as the most relevant. To ensure a proper understanding of the meaning of the topics, a table for each sample was added to the Appendix, containing all 10 description words, not only the first four.

For a better understanding of the 120 topics, five categories were created and each topic has been assigned to one or, in some cases, two of them. The division of the topics into categories is presented in Table 3, along with an overview of the percentages of topics that fall into each category. Because of the few topic exceptions that cannot be associated with a single category but with two, the percentages add up to more than 100%, which means they cannot be interpreted as probabilities. The five categories are:

1. Election, Trump, Biden, Fraud,
2. Parler,
3. Pandemic,
4. Religion,
5. Others (User Agreement, Celebrations, Education, etc.).

	Topics_Complete_Sample	Topics_November_Sample	Topics_December_Sample	Topics_January_Sample
-1	not_trump_people_like	not_trump_get_like	not_people_trump_get	not_people_trump_like
0	meeting_joined_here_everyone	parler_tos_parlersupport_tip	good_lol_awesome_nice	parler_echo_welcome_gab
1	alive_face_mixing_constant	alive_face_mixing_constant	yes_absolutely_idea_interesting	absolutely_sad_america_pig
2	parler_tos_parlersupport_tip	meeting_joined_everyone_here	parler_welcome_looking_meeting	trump_president_never_nope
3	clicking_below_text_link	below_clicking_text_link	treason_hang_gitmo_rope	pray_praying_wow_prayer
4	commentary_favorite_follow_welcome	commentary_favorite_follow_you	texas_court_supreme_scotus	shot_evil_gun_woman
5	commentary_favorite_follow_great	commentary_favorite_follow_you	bless_news_god_fake	lol_beautiful_sure_funny
6	fox_news_cnn_tucker	thank_thanks_hello_hey	love_this_guy_president	penny_mike_coward_juda
7	happy_awesome_thanksgiving_new	fox_news_newsmax_done	true_sad_disgusting_american	china_chinese_virus_biden
8	exactly_done_happen_trump	lol_absolutely_yep_lmao	exactly_crook_dems_happen	exactly_plan_trust_rudy
9	thanks_this_watch_bullshit	true_idiot_moron_vote	amen_brother_praise_god	bless_god_yep_dan
10	treason_jail_surprise_prison	facebook_twitter_censorship_google	barr_durham_doj_bill	agreed_awesome_antifa_aoc
11	yes_hell_mean_yeah	love_president_disgusting_lady	never_shit_piece_asshole	amen_brother_awoman_hallelujah
12	thank_hello_you_parler	georgia_recount_senate_audit	mask_wear_wearing_not	supporter_antifa_maga_trump
13	amen_brother_name_praise	news_nice_fake_idea	joe_biden_concede_china	capitol_storm_capital_police
14	georgia_kemp_vote_recount	awesome_beautiful_wrong_omg	praying_pray_prayer_unity	twitter_suspended_account_deleted
15	echo_twitter_suspended_fakebook	china_kraken_virus_release	beautiful_history_best_president	fraud_election_voter_fraudulent
16	praying_pray_prayer_correct	praying_prayer_pray_god	christmas_merry_family_melania	thank_you_spot_sir
17	police_cop_shot_veteran	exactly_stupid_late_democrat	jail_prison_criminal_crooked	pelosi_nancy_schumer_chuck
18	wow_evil_snake_ugly	maga_content_truly_supporter	thank_thanks_you_matt	meeting_joined_forward_looking
19	fraud_voter_election_proof	enjoy_truefreespeech_parlerconcierge_interact	covid_bullshit_death_test	video_youtube_link_watch
20	stand_matt_safe_shut	fraud_voter_election_evidence	swamp_rat_basement_drain	apple_app_google_phone
21	covid_flu_death_test	happy_thanksgiving_christmas_birthday	senator_cruz_ted_senate	yes_did_course_pathetic
22	true_hero_revolution_please	tucker_hannity_fox_laura	yep_corruption_hole_shocker	fight_concede_transition_fighting
23	pelosi_nancy_schumer_crazy	amen_leo_that_glory	vaccine_nope_take_not	traitor_list_deserved_deserves
24	texas_court_agreed_supreme	judge_pennsylvania_deleted_boom	pelosi_nancy_ice_cream	georgia_abrams_vote_kemp
25	sad_wtf_unbelievable_country	ballot_vote_mail_count	school_teacher_kid_education	steal_election_dems_stolen
26	love_scott_heart_president	yes_time_damn_straight	fuck_pay_paid_bingo	happy_new_year_birthday
27	fight_fighting_believe_keep	wow_interesting_got_ready	video_rumble_youtube_share	facebook_twitter_thanks_parler
28	yep_good_question_one	never_nope_surprised_penny	obama_mike_clinton_hillary	love_snake_congratulation_president
29	enjoy_hope_truefreespeech_parlerconcierge	obama_mike_history_worst	surprise_know_what_knew	senator_cruz_ted_senate

Table 2: Topic descriptions

	Complete Sample	November Sample	December Sample	January Sample
1. Election, Trump, Biden, Fraud				
Topic Indices	3, 6, 8, 10, 11, 14, 16, 17, 19, 20, 22, 23, 24, 25, 26, 27	3, 4, 5, 7, 9, 10, 11, 12, 13, 15, 16, 17, 18, 20, 22, 23, 24, 25, 28, 29	1, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16, 17, 18, 20, 21, 22, 24, 27, 28	1, 2, 4, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 19, 20, 22, 23, 24, 25, 27, 28, 29
Count	16	20	22	24
Percentage	53.33%	66.67%	73.33%	80.00%
2. Parler				
Topic Indices	0, 1, 2, 3, 4, 5, 12, 15, 29	0, 1, 2, 4, 5, 6, 10, 11, 19	2, 27	0, 14, 18, 20, 27
Count	9	9	2	5
Percentage	30.00%	30.00%	6.67%	16.67%
3. Pandemic				
Topic Indices	21	15	12, 19, 23	7
Count	1	1	3	1
Percentage	3.33%	3.33%	10.00%	3.33%
4. Religion				
Topic Indices	13, 16	16	9, 14	3
Count	2	1	2	1
Percentage	6.67%	3.33%	6.67%	3.33%
5. Others (User Agreement, Celebrations, Education etc.)				
Topic Indices	7, 9, 18, 28	8, 14, 21, 26, 27	0, 1, 25, 26, 29	5, 21, 26
Count	4	5	5	3
Percentage	13.33%	16.67%	16.67%	10.00%

Table 3: Topic categories

In each of the four data samples, at least 50% of the topics are addressing the 2020 U.S. presidential election. Beginning with a percentage of almost 67% of Parler discussions in November 2020 related to the election, the amount increases each month, with more than 73% of the topics in December 2020 and 80% of the ones in January 2021 discussing the presidential battle.

In November 2020 the chats referred mostly to the American multinational conservative television channel Fox News (topic 7 mentions the news anchor Maria Bartiromo and topic 22 Tucker Carlson, Sean Hannity, Laura Ingraham, and Mark Levin, all Fox News hosts supporting Trump), and to the alleged election fraud in some states (while topic 20 is referring to British prime minister Boris Johnson, topic 12 is drawing attention to Georgia's Incumbent Governor Brian Kemp and topic 24 to the election conspiracist Court judge Jeanine Pirro). Most of the discussions seemed to support Donal Trump (topic 13 refers to him as a "good source") and to blame the democrats for the problems in the U.S. (topic 17 describes the democrats as "dumb" and "stupid" and topic 29 mentions the "worse" "history" and portrays former first lady Michelle and former President Barack Obama "narcissists" who are an "embarrassment").

The discussions in December 2020 are also to a large extent debating the presidential elections. In line with the third topic which expresses the opinion that the "traitors" should be sentenced by the "tribunal" with "hanging" or sent to the "gitmo" (Guantanamo Bay Detention Camp), topic 17 calls California's democratic governor Gavin Newsom a "criminal" who shall go to "prison". The fourth topic is related to the "Texas v. Pennsylvania" lawsuit filed on December 8, 2020, at the U.S. Supreme Court contesting the administration of the 2020 presidential election in certain states (*Texas v. Pennsylvania - Wikipedia*, 2022). Topics 10 and 20 refer to the former attorney general William Barr as a "snake" or "rat" because he considered Trump's voter fraud allegations lies, although he was working for his administration (*William Barr: no evidence of voter fraud that would change election outcome*, 2021). Donald Trump is not the only

one appreciated in Parler discussions (topic 7 is praising him for giving "color" to the "disgusting America"), but also the first lady Melania Trump (topic 16) alongside all his political supporters are esteemed (topic 21 "thanks" Republican senators Ted Cruz and Josh Hawley for supporting the false claims of voter fraud).

The favorite conversation subject for January's 2021 discussion on Parler is clearly the election too. Apart from the usual fraud claims (topic 15 and 25), the appreciation shown towards former president Donald Trump and his supporters (topics 28 and 29 and topic 8 encouraging to "trust" the "plan" set up by the republican lawyer Rudy Giuliani), and the hate received by democrats (topic 10 describing the democrat U.S. Representative Alexandria Ocasio-Cortez an "excellent target" and topic 17 referring to Nancy Pelosi, the democratic speaker of U.S. House gross, as a "witch"), the focus of the discussions is set on the events taking place in Washington. The users of the platform express their disappointment about the official and irreversible loss of Donald Trump in favor of Joe Biden (topic 1 describes them feeling "absolutely sad" about "America", which underwent a "disgusting joke" that was "definitely planned"). The fourth most common conversation topic brings up Ashli Babbitt, the woman who was shot by a police officer during the Capitol riot and the users consider that the "evil cop" should be "fired " because she was "unarmed". Topics 12 and 13 discuss the events of the "capitol storm", mentioning "ANTIFA", the left-wing anti-fascist and anti-racist political movement in the U.S., the "MAGA" ("Make America Great Again") movement, and Trump's "supporter" "rally". Trump's Vice President Mike Pence, also the President of the Senate, who declared Joe Biden the official winning candidate on January 7, 2021, is called a "coward", "traitor" and "juda" (topic 6), who "deserves shame" (topic 23). While the link to Trump's concession video posted on 7 January 2021 is shared across the platform (topic 19), the users consider that he will "never" truly "concede", stop to "fight", and allow for a "smooth" "transition" (topic 22).

The topics related to the elections found by the algorithm in the complete sample only reiterate all discussion subjects explained above. The fraudulent vote (topics 10, 14, 19, 24, and 27), the disagreement with the opposition (topics 23 and 25), and the approval of Donald Trump's supporters, both on television and politics (topics 6, 8, 22, 27), as well as the U.S. Capitol riots (17) are the most prominent.

The second most notable category of topics in both the November sample and the sample containing parleys from all of the three months was the category referring to Parler. Around 30% of the total subjects bring up the platform itself. For example, in the complete sample, topics 0, 4, 5, and 12 consist of common welcoming words posted on social media networks when welcoming new users ("commentary", "favorite", "follow", "welcome", "hello"). The broad interpretation of free speech, the signature attribute of Parler, is also mentioned, with topic 1 describing the users feeling "glad" that the platform is "keeping speech" "actually free" and topic 29 speaking of the "found freedom". Topic 15 refers to Twitter and Facebook as "asshole ", both filtering out posts on their platforms, and portrays the anger Parler users experience because of their account being "suspended" by the bigger platforms.

The next topic categories are the pandemic and religion. These are less significant in comparison with the first two (elections and Parler), although at least one topic from each sample refers to the COVID-19 virus and at least one to religious beliefs.

Parler users seem to have been the most concerned about the virus in December 2020, with three topics being related to the pandemic (topics 12, 19 and 23). Topic 12 clearly refers to the Parler users' disapproval of the measures taken by the U.S. to stop the spread of the pandemic, mentioning them "not" willing to "wear" a "mask" and comparing the coronavirus disease to a "flu". Topic 19 illustrates the pandemic as a "bullshit", leading to the supposition that some of Parler users do not believe in the existence of the virus. For the description of topic 23, the algorithm uses the following 10 words: "vaccine nope take not speed mine taking animal immunity never", hinting at the discussion expressing strong opposition to the vaccine.

Most of the topics related to religion are also referring to former president Donald Trump. Topic 16 from the complete sample is mentioning a "prayer" for the "president" and "courage" for "family trump". The same topic number from the November dataset includes the following words "god bless president". In the December sample topic 14 is referring to a "prayer" for "unity" and "recovery" and to the "brave" "president" "Trump". Interestingly, the family of Trump is also mentioned, Ivanka Trump, the daughter of the president being called a "sister" by Parler users in topic 9 from the December dataset.

The last category includes topics appearing in just a fragment of the samples. These were related to celebrations, education, or showing the users' agreement. Some of them seemed irrelevant for the analysis or their interpretation proved to be challenging.

For the November sample, topics 8 ("lol absolutely yep"), 26 ("yes time damn straight"), and 27 ("wow interesting got ready definitely always right") prove that a significant part of Parler users have the same views. Topic 21 ("happy thanksgiving christmas birthday veteran family cancel marine day kiss)" is related to the celebrations and mentions U.S. Marine Corps who received Thanksgiving meals after the annual Adopt-a-Marine program was canceled due to the pandemic (*1,000 Marines receive Thanksgiving meals after pandemic forces cancellation of Adopt-a-Marine program*, 2020). While topic 25 in the December sample is clearly related to education ("school teacher kid education child union student teaching college parent"), topics 26 ("fuck pay paid bingo money piss bought off sickening pissed") and 29 ("surprise know what knew now say something that sweet and") are not intelligible. Topic 1, consisting of the following words "yes absolutely idea interesting course perfect", shows that a majority approves the opinions of the other users on the platform.

The same is valid for the January sample, with topics 5 ("sure", "right") and 21 ("yes did course pathetic") displaying the same views of the people active on Parler. Topic 26 is related to the recent celebrations, Christmas and New Year.

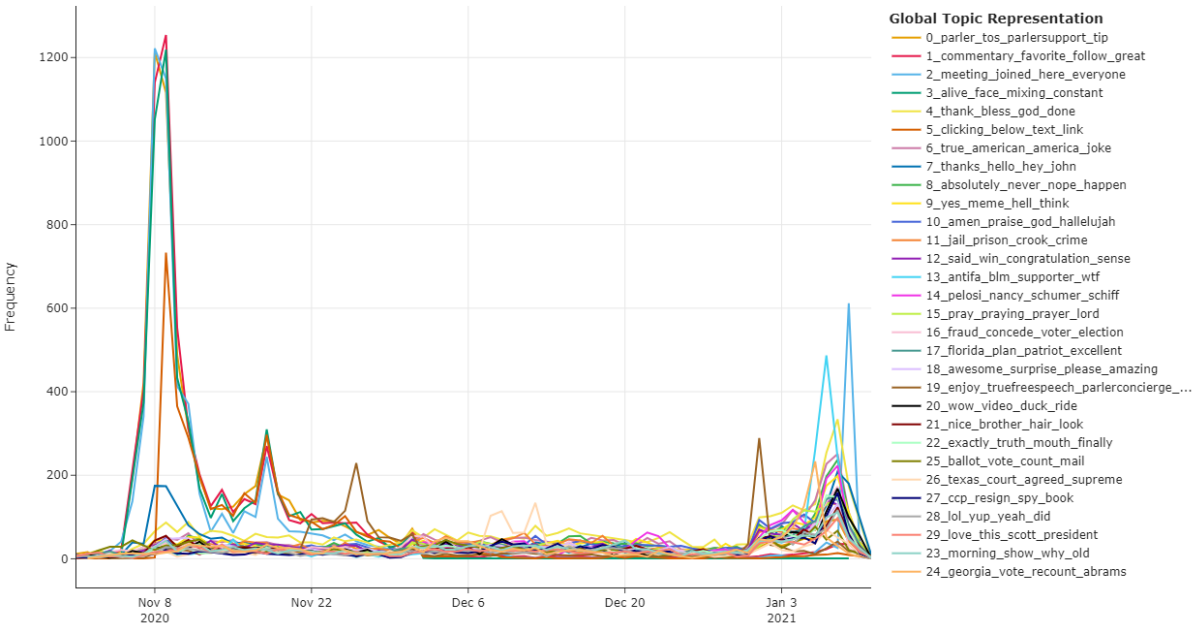
4.2 Dynamic Topic Modelling

Not only the actual topics discovered by the algorithm in the Parler samples are significant, but also their evolution over time. When taking the temporal dimension into account, BERTopic also solves the dynamic topic modelling (DTM) task. However, because of the stochastic nature of UMAP, as explained in the Methodology section, DTM BERTopic models found slightly different topics in comparison with the ones computed by the models above, although they received the same data samples as input. Still, the meaning of the topics calculated by the algorithms does not change, variations occur only in the order of

the topics and chosen words in the descriptions.

While Figure 3 displays the overall trends derived from the complete sample, for the November, December, and January samples DTM results can be found in the Appendix. As shown in the figure, the discussions on Parler focused mainly on a specific subject in the first half of November 2020 and at the beginning of January 2021. Topics 0 ("parler parlersupport tip"), 1 ("commentary favorite follow"), 2 ("meeting joined here everyone"), and 5 ("clicking below text link") are clearly related to Parler welcoming new users on the platform. The huge increase takes place around the general election day, November 3, 2020, and discussions about Parler seem to stay on a higher level until the end of the month. Topic 19 mentions "enjoy truefreespeech", highlighting the unrestricted freedom of speech on the platform, especially on two occasions: "thanksgiving" and "new year". The second increase of discussions volume took place around the Electoral College vote count from January 6, 2021, when the rioters broke into the Capitol. Topic 13 mentions the left-wing anti-fascist and anti-racist political movement "antifa" and the black Lives Matter movement ("blm"), as well as the "supporters" of "trump". Topic 6 is related to the attack on the Capitol building, bringing up Ashli Babbitt, one of the rioters who died during the insurrection, and describes "America" as a 'joke". Topic 16 is also among the most intensively discussed, suggesting that Trump will not concede ("concede course not"), while also highlighting the alleged "voter fraud" during the "election".

Figure 3: Dynamic Topic Modelling for the complete sample



5 Discussion

5.1 Results Summary

As described in the Results section, the vast majority of the topics found by the algorithm are related to the 2020 U.S. presidential election. Although each sample has a slightly different focus, depending on the actual election timeline, subjects related to the former President Donald Trump, his election rival Joe Biden, the alleged voter fraud, the Capitol attack, and to numerous Democratic and Republican politicians, are common. In addition to the topics discussing the presidential battle between Trump and Biden, a few are also referring to the platform itself. Most of these topics praise Parler for not restricting the freedom of speech and disagree with other social networks such as Twitter and Facebook which limit content and ban users. Another topic category is related to the COVID-19 pandemic, with discussions mainly expressing opposition regarding the vaccine and the measures taken to stop the spread of the virus. Many users on the platform seem to consider the coronavirus disease a "flu" or even an invention. Other secondary topics are religion, the education system, and the celebrations taking place in November and December: Thanksgiving, Christmas, and New Year.

5.2 Research Questions Answers

The first research question drew attention to the users of the social network Parler and to their discussion subjects. Based on the topics computed by the algorithm, the suppositions made by researchers like M. Otala et al. (2021) and Aliapoulios et. al 2021 are now backed up by data. The hypothesis that the user base of Parler contains a substantial number of accounts created and used by conservatives, Trump supporters, patriots, and religious persons has been proven right. However, the results do not imply that Parler users are only people with strong right-wing political opinions, extremists, or conspiracy theorists. Assuming that users of the platform belong exclusively to those banned from bigger social networks, only because of certain Parler discussions condemning the restriction of the freedom of speech on Twitter or Facebook, is unfounded. Still, the statement that people who have felt restricted on other platforms are active on Parler is supported by some fanatical discussion topics and the way the rather radical viewpoints are expressed.

Two similarities have been observed in the majority of the conversation subjects. First, for a considerable number of topics, offensive words were selected by the algorithm as most appropriate for their description. Ranging from mild swear words to obscenities, this kind of language is inadequate even for an informal context. Second, the perspectives of the people who took part in the discussions were remarkably similar. Users often seemed to agree with the other opinions on the platform. Both remarks raise alarms, as they lead to the assumption that Parler users are a group of like-minded people who support each other and offend others.

The second research questions ask for the investigation of the relation between the constitutional freedom of speech and the topics discussed on Parler. Expressing opinions unrestrained, although a freedom guaranteed by the First Amendment, does not come without limits, as explained in the Theoretical Foundation section. Some of the discussions on Parler seem to include advocacy of illegal conduct, fighting words, true threats,

defamation, and false statements. Especially in the context of the Capitol riots, Parler users seem to have incited the illegal attack on the building, communicating serious threats of physical injury to both the protesters gathered in Washington and the Senators inside the Capitol. False statements have not only been made about several political figures associated with the Democratic Party, in an attempt to damage their reputation, but also about the COVID-19 virus, regarded as a "lie". These kinds of expression and conspiracies are not protected under the First Amendment.

Regarding the third research question, the findings prove that Parler discussions are connected to the 2020 U.S. presidential elections and the Capitol riots from January 6, 2021. Not only were the discussions on the platform debating the events of the elections and most intensively on key dates, but notable increases of new users were also witnessed, especially in November 2020 and January 2021. However, making a statement about the direction of this relationship turns out to be a challenging task. Evidence of an influence of the topics discussed on Parler on the actual events in Washington is unclear. Allegations of voter fraud, conspiracy theories, and political inquiries were common discussion subjects, but a topic related to the actual planning of a riot on the Capitol was not discovered by the algorithm. However, the aftermath of the attack involving the death and injury of many people, the recognition of Joe Biden's win, and Donald Trump's concession speech, were highly debated. Parler users express their anger towards the police officer who fatally shot Ashli Babbitt, one of the rioters, and are not willing to believe their preferred election candidate, Donald Trump, actually lost.

5.3 Strengths and Limitations

The results discussed above highlight the great strength of this thesis: providing objective evidence. By using a data-driven approach and neural topic modelling, the findings are almost entirely unbiased and enable a trustworthy analysis. The assumptions made about Parler in the context of the 2020 U.S. presidential elections were proved to be to a large extent right. For example, the results point out that the users of the platform have extremist views. In addition to that, the connection between Parler as a social media platform, freedom of speech, and real-life events was explained. Nonetheless, time, storage, computational constraints, as well as dataset issues are limitations of this research. Because the parleys were only collected from a random sample of four million users, the dataset may not necessarily offer an accurate representation of the topics discussed, Aliapoulios et. al 2021 acknowledge. Apart from that, the analysis of this thesis has only been conducted on four samples of posts and comments from Parler, which may further decrease the generalization power of the findings.

6 Conclusion

The aim of this bachelor's thesis was to inspect what impact discussions on social platforms can have on real life. The way society is affected by a rather broad interpretation of the freedom of speech guaranteed by the First Amendment was also observed. By analyzing posts and comments from the Parler social network in the context of the 2020 U.S. presidential election, this research connected the discussions between the people active on the platform and the election timeline. The period taken into consideration was November 2020 to January 2021, which included the most relevant election events: the general election day, the vote of the electors in their States, and the day the Congress met in joint session to count the electoral votes. To answer the three research questions, BERTopic, an algorithm implementing the state-of-the-art BERT model for simple and dynamic topic modelling, was performed on four samples from a public dataset of Parler posts and comments. Before being forwarded as input in the algorithms, the samples were thoroughly preprocessed.

The results revealed that between 66% and 80% of the conversations on the platform were related to the elections. Parler users were linked to Trump supporters, patriots, and conspiracists, often expressing their extremist views using vulgar language. For example, political democratic personalities and rivals of the former president Donald Trump, received hate online, being called "stupid", "assholes", or "piece" of "shit". In contrast, Trump and his whole family, conservative television hosts, as well as lawyers publicly advocating for the false allegation of voter fraud, were highly regarded by the users. The former president was described as "brave" by people active on the platform, who consider him a "good source".

Apart from the discussions about the presidential election, the second most prominent category of topics was related to Parler along with other social media networks like Twitter and Facebook. While those two were blamed for filtering out posts and banning users, Parler was appreciated for allowing opinions regardless of the content.

The third most common subject on Parler was the COVID-19 pandemic. The users clearly expressed their disagreement with the restrictions and the vaccine. Many considered the virus just a "flu" or even a lie, which proves that conspiracy theories were spread and supported across the platform.

Moreover, the results of BERTopic for dynamic topic modelling showed that Parler registered an increase in the number of users both in November 2020 and January 2021. By taking the temporal dimension into account, the algorithm discovered that conversations related to Parler and the welcoming of new users were more frequent after the general election day and around the U.S. Capitol riots. The restrictions imposed by bigger social media networks to stop misinformation about the elections and voter fraud could have affected people to such a great extent that they decided to switch to Parler.

The findings of this research could not only raise awareness about the serious consequences speech can have on society and people's lives, but also help both government authorities and private institutions make knowledgeable decisions regarding their policies.

A Appendix

As explained in the Results section, each topic computed by the BERTopic algorithm was described using 10 words. In some cases the meaning of the topic can only be properly understood after analyzing the entire description, not just the first four words presented in Table 2. Therefore, Tables 4, 5, 6, and 7 display the complete descriptions for each of the four samples in the following order: complete sample, November 2020, December 2020, and January 2021.

Figures 4, 5 and 6 show the evolution of topics over time, calculated by the BERTopic algorithm for dynamic topic modelling for the November 2020, December 2020, and January 2021 samples.

Figure 4: Dynamic Topic Modelling for the November 2020 sample

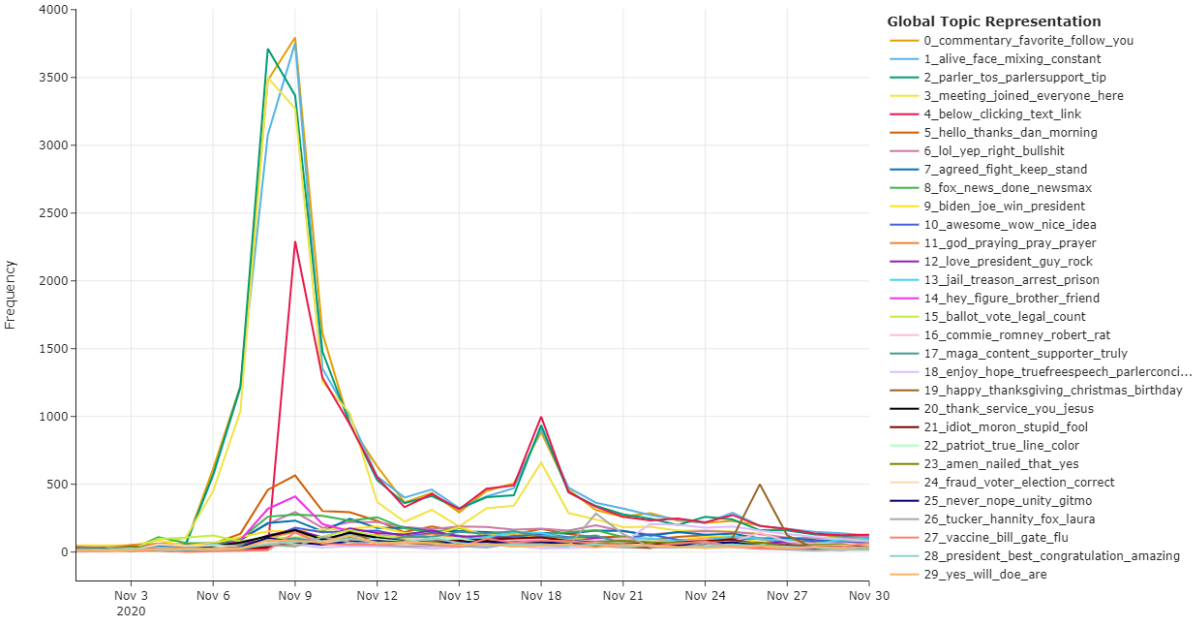


Figure 5: Dynamic Topic Modelling for the December 2020 sample

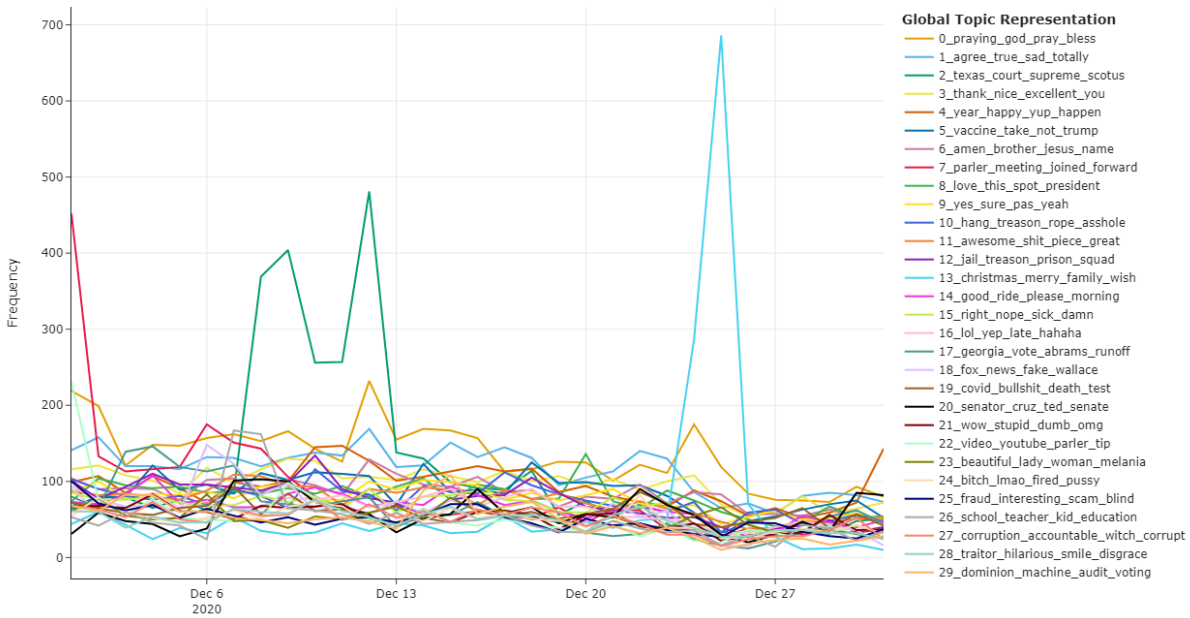
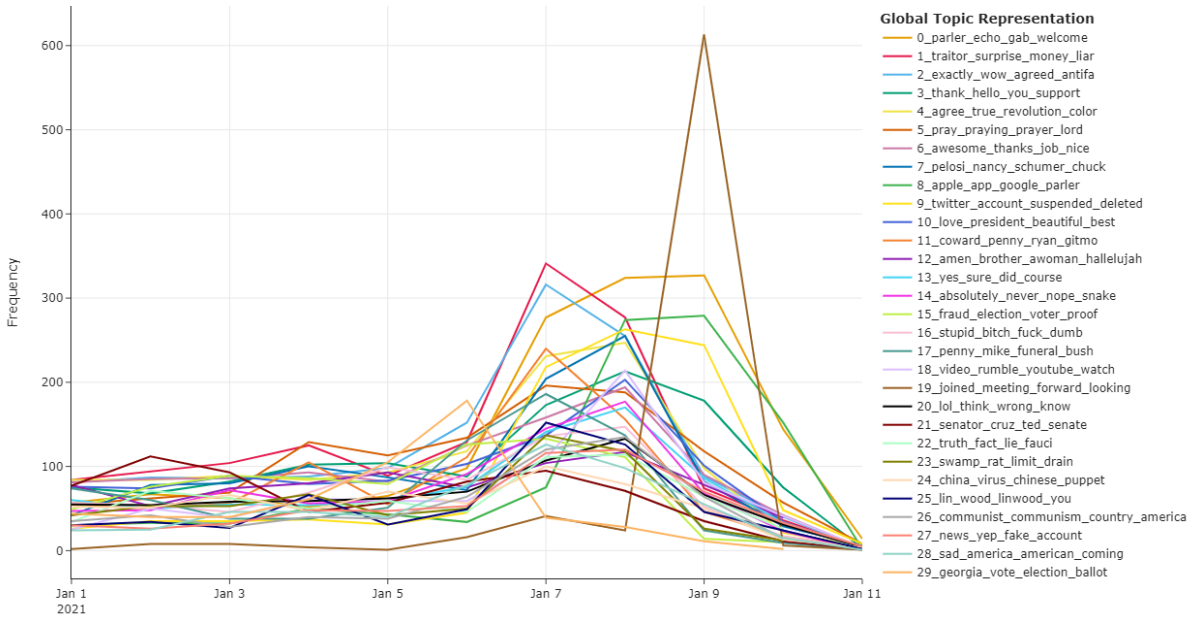


Figure 6: Dynamic Topic Modelling for the January 2021 sample



Nr	Word 1	Word 2	Word 3	Word 4	Word 5	Word 6	Word 7	Word 8	Word 9	Word 10
-1	not	trump	people	like	get	need	would	president	one	know
0	meeting	joined	here	everyone	forward	looking	parler	trip	right	lookey
1	alive	face	mixing	constant	bias	keeping	speech	glad	actually	free
2	parler	tos	parlersupport	tip	channel	data	youtube	page	check	place
3	clicking	below	text	link	help	sure	america	make	great	welcome
4	commentary	favorite	follow	welcome	great	you	yay	bingo	following	nothing
5	commentary	favorite	follow	great	you	welcome	what	bahahaha	bahahahaha	going
6	fox	news	cnn	tucker	newsmax	paul	hannity	watch	ryan	watching
7	happy	awesome	thanksgiving	new	news	year	idea	fake	birthday	excellent
8	exactly	done	happen	trump	going	president	luck	coming	knew	nothing
9	thanks	this	watch	bullshit	got	bull	haha	did	away	sharing
10	treason	jail	surprise	prison	surprised	accountable	charge	treasonous	scumbag	not
11	yes	hell	mean	yeah	sure	trump	lunatic	cheer	say	truck
12	thank	hello	you	parler	figure	sir	jim	trying	service	thanks
13	amen	brother	name	praise	jesus	sister	psalm	change	that	god
14	georgia	kemp	vote	recount	abrams	election	runoff	senate	ballot	republican
15	echo	twitter	suspended	fakebook	asshole	twatter	back	suck	ditto	dick
16	praying	pray	prayer	correct	linwood	president	god	courage	family	trump
17	police	cop	shot	veteran	unarmed	defund	officer	fired	ashli	capitol
18	wow	evil	snake	ugly	amazing	omg	demon	smile	teeth	bear
19	fraud	voter	election	proof	fraudulent	biden	not	glitch	vote	caught
20	stand	matt	safe	shut	target	perfect	stay	president	weak	strong
21	covid	flu	death	test	positive	doctor	hospital	scam	not	suicide
22	TRUE	hero	revolution	please	color	real	american	story	nigger	hope
23	pelosi	nancy	schumer	crazy	ice	schiff	cream	chuck	bat	wine
24	texas	court	agreed	supreme	case	scotus	lawsuit	state	candace	texan
25	sad	wtf	unbelievable	country	damn	definitely	straight	take	back	true
26	love	scott	heart	president	break	nail	breaking	terrence	tear	guy
27	fight	fighting	believe	keep	rock	back	president	trump	not	freedom
28	yep	good	question	one	cute	look	brilliant	wth	that	buddy
29	enjoy	hope	truefreespeech	parlerconcierge	newuser	interact	fun	found	freedom	new

Table 4: Complete descriptions of the topics from the complete sample

Nr	Word 1	Word 2	Word 3	Word 4	Word 5	Word 6	Word 7	Word 8	Word 9	Word 10
-1	not	trump	get	like	need	would	biden	election	president	know
0	parler	tos	parlersupport	tip	channel	data	page	youtube	check	place
1	alive	face	mixing	constant	bias	keeping	speech	glad	actually	free
2	meeting	joined	everyone	here	forward	looking	parler	and	guaranteed	also
3	below	clicking	text	link	help	sure	america	make	trump	great
4	commentary	favorite	follow	you	great	welcome	kevins	kevinslack	deja	rsbnetwork
5	commentary	favorite	follow	you	great	welcome	yay	ccp	name	remember
6	thank	thanks	hello	hey	figure	hang	aboard	you	trying	service
7	fox	news	newsmax	done	watch	dead	oan	fuck	watching	maria
8	lol	absolutely	yep	lmao	yup	work	yeah	door	what	hahaha
9	true	idiot	moron	vote	voted	fool	pig	hilarious	omar	biden
10	facebook	twitter	censorship	google	blocked	duck	censored	not	banned	zuckerberg
11	love	president	disgusting	lady	best	nut	rock	dorsey	jack	perfect
12	georgia	recount	senate	audit	kemp	vote	ballot	state	not	need
13	news	nice	fake	idea	excellent	account	read	djt	good	source
14	awesome	beautiful	wrong	omg	show	amazing	great	disgrace	look	fantastic
15	china	kraken	virus	release	chinese	test	biden	beijing	lou	soul
16	praying	prayer	pray	god	bless	president	sir	family	thank	trump
17	exactly	stupid	late	democrat	dumb	thought	little	abortion	girl	hawaii
18	maga	content	truly	supporter	part	million	rally	trump	follow	glad
19	enjoy	truefreespeech	parlerconcierge	interact	newuser	fun	found	freedom	new	hope
20	fraud	voter	election	evidence	not	fraudulent	scam	boris	massive	exposed
21	happy	thanksgiving	christmas	birthday	veteran	family	cancel	marine	day	kiss
22	tucker	hannity	fox	laura	mark	levin	bye	news	carlson	watch
23	amen	leo	that	glory	god	yes	terrell	love	lord	thank
24	judge	pennsylvania	deleted	boom	real	jeanine	whistleblower	state	usps	account
25	ballot	vote	mail	count	paper	counted	correct	counting	watermark	not
26	yes	time	damn	straight	sweet	fly	flying	airline	doe	plane
27	wow	interesting	got	ready	definitely	say	get	always	right	retard
28	never	nope	surprised	penny	president	happen	not	give	going	wth
29	obama	mike	history	worst	michelle	michael	night	barry	embarrassment	narcissist

Table 5: Complete descriptions of the topics from the November 2020 sample

Nr	Word 1	Word 2	Word 3	Word 4	Word 5	Word 6	Word 7	Word 8	Word 9	Word 10
-1	not	people	trump	get	need	like	would	election	one	know
0	good	lol	awesome	nice	lmao	bitch	morning	surprised	point	omg
1	yes	absolutely	idea	interesting	course	perfect	did	trump	finally	let
2	parler	welcome	looking	meeting	joined	forward	here	everyone	enjoy	massupvote
3	treason	hang	gitmo	rope	treasonous	squad	firing	tribunal	traitor	hanging
4	texas	court	supreme	scotus	case	state	lawsuit	join	not	suit
5	bless	news	god	fake	great	hero	excellent	question	congratulation	gift
6	love	this	guy	president	spot	got	kidding	man	popcorn	like
7	true	sad	disgusting	american	america	gross	soon	coming	color	djt
8	exactly	crook	dems	happen	accountable	corrupt	held	way	shame	democrat
9	amen	brother	praise	god	jesus	name	ivanka	sister	lord	help
10	barr	durham	doj	bill	fired	resign	fat	fraud	not	investigation
11	never	shit	piece	asshole	bye	crazy	give	yeah	president	bail
12	mask	wear	wearing	not	flu	diaper	work	virus	people	subpoena
13	joe	biden	concede	china	door	beijing	not	feather	sleep	sleepy
14	praying	pray	prayer	unity	recovery	god	president	brave	speedy	trump
15	beautiful	history	best	president	ever	family	lady	greatest	definitely	statue
16	christmas	merry	family	melania	bless	god	nailed	smile	president	lady
17	jail	prison	criminal	crooked	prize	peace	creep	prosecuted	gavin	need
18	thank	thanks	you	matt	sir	service	patriot	already	god	president
19	covid	bullshit	death	test	positive	flag	false	relief	died	dead
20	swamp	rat	basement	drain	snake	creature	part	barr	north	trump
21	senator	cruz	ted	senate	hawley	thank	johnson	wipe	missouri	house
22	yep	corruption	hole	shocker	orange	county	deep	cult	wth	brennan
23	vaccine	nope	take	not	speed	mine	taking	animal	immunity	never
24	pelosi	nancy	ice	cream	schumer	ben	jerry	nasty	end	chuck
25	school	teacher	kid	education	child	union	student	teaching	college	parent
26	fuck	pay	paid	bingo	money	piss	bought	off	sickening	pissed
27	video	rumble	youtube	share	sharing	camera	song	shocked	shocking	shared
28	obama	mike	clinton	hillary	michael	michelle	barack	obamas	suicide	narcissist
29	surprise	know	what	knew	now	say	something	that	sweet	and

Table 6: Complete descriptions of the topics from the December 2020 sample

Nr	Word 1	Word 2	Word 3	Word 4	Word 5	Word 6	Word 7	Word 8	Word 9	Word 10
-1	not	people	trump	like	get	need	would	one	know	president
0	parler	echo	welcome	gab	hello	fakebook	post	join	glad	hey
1	absolutely	sad	america	pig	disgusting	joke	american	definitely	planned	save
2	trump	president	never	nope	win	going	not	lot	will	happen
3	pray	praying	wow	prayer	lord	god	family	safe	mercy	father
4	shot	evil	gun	woman	unarmed	ashli	blood	gitmo	fired	cop
5	lol	beautiful	sure	funny	right	nice	know	sense	lmao	hilarious
6	penny	mike	coward	juda	traitor	funeral	tom	vice	president	trump
7	china	chinese	virus	biden	beijing	joe	sold	communist	sell	stock
8	exactly	plan	trust	rudy	what	barr	bring	bingo	thought	boom
9	bless	god	yep	dan	president	best	thank	you	ever	forget
10	agreed	awesome	antifa	aoc	idea	please	excellent	target	report	great
11	amen	brother	awoman	hallelujah	awomen	jesus	nailed	sister	trial	amoron
12	supporter	antifa	maga	trump	loser	surprise	blame	not	hat	dressed
13	capitol	storm	capital	police	antifa	video	building	staged	bus	rally
14	twitter	suspended	account	deleted	badge	tweet	delete	jack	deactivated	purge
15	fraud	election	voter	fraudulent	evidence	proof	fauci	not	vote	investigation
16	thank	you	spot	sir	ditto	information	update	president	thanks	patriot
17	pelosi	nancy	schumer	chuck	witch	cream	ice	pelosis	speaker	gross
18	meeting	joined	forward	looking	here	everyone	parler	coming	saw	cute
19	video	youtube	link	watch	concession	post	watching	channel	source	open
20	apple	app	google	phone	parler	duck	store	android	browser	iphone
21	yes	did	course	pathetic	already	are	milo	many	not	sure
22	fight	concede	transition	fighting	conceded	not	trump	never	smooth	said
23	traitor	list	deserved	deserves	shame	remember	karma	cheney	well	deserve
24	georgia	abrams	vote	kemp	election	stacy	stacey	georgian	runoff	state
25	steal	election	dems	stolen	ball	hahaha	stole	stealing	insanity	landslide
26	happy	new	year	birthday	radio	ear	christmas	god	cheer	bless
27	facebook	twitter	thanks	parler	rumble	banned	got	instagram	account	ban
28	love	snake	congratulation	president	this	ride	horse	great	trump	devin
29	senator	cruz	ted	senate	hawley	josh	thank	house	stand	republican

Table 7: Complete descriptions of the topics from the January 2021 sample

B List of Abbreviations

AI Artificial Intelligence

$BERT$ Bidirectional Encoder Representations from Transformers

$BPTM$ Bayesian Probabilistic Topic Model

$c-TF-IDF$ class-based Term Frequency - Inverse Document frequency

DTM Dynamic Topic Modelling

$HDBSCAN$ Hierarchical Density-Based Spatial Clustering of Applications with Noise

LDA Latent Dirichlet Allocation

NLP Natural Language Processing

NTM Neural Topic Model

$U.S.$ United States

$UMAP$ Uniform Manifold Approximation and Projection

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